

claude-windows / 9f6357d8-0e35-491a-a830-052e094c7f37

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_b6ef6275-de5\agent-a535589560849586c.jsonl`  
- SHA-256: `11502c709b897ae48476d2471c12e8af5f2a5750497294456e65377e06567950`  
- Source modified: `2026-06-21T13:58:08+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_b6ef6275-de5`  
- session_id: `9f6357d8-0e35-491a-a830-052e094c7f37`
```

Transcript

1. user (2026-06-21T13:56:13.269Z)

Read the LOCALIZATION + non-tech UX track docs and extract status + next steps. Read khelsutra (C:/Users/avidu/Projects/khelsutra) docs/FRIENDLY_AND_MULTILINGUAL.md (the \$7 deliverables/progress tracker ? note which B*/A* rows are ? vs ? vs ?, and the \$9 DoD), and the engine (C:/Users/avidu/Projects/badminton-highlight-indexer) design docs/I18N_PLAN.md + docs/UI_PROPOSAL.md. The SPA (web/) is fully en+kn+hi already; capture what REMAINS (e.g. B3 site/ t() shim, A-items like WhatsApp share / mobile touch targets / friendly error mapping, native-speaker review of AI-drafted kn/hi, engine P2 backend localization). Tag each gated/effort. Return structured JSON (track="localization+UX").

2. assistant (2026-06-21T13:56:15.728Z)

I'll start by pulling the latest on both repos, then read the localization + UX track docs.

3. user (2026-06-21T13:56:21.265Z)

fatal: Cannot fast-forward to multiple branches.
---KHELSUTRA---
Already up to date.

4. user (2026-06-21T13:56:21.540Z)

```
ANNOTATE_AND_TRAIN.md  
DRY_RUN_CUJ.md  
FRIENDLY_AND_MULTILINGUAL.md  
LOCAL_APPLIANCE_ARCHITECTURE.md  
PER_RALLY_SELECTION.md  
PORTABILITY.md  
README.md  
REQUEST_ACCESS_FORM.md  
---ENGINE DOCS---  
I18N_PLAN.md  
UI_PROPOSAL.md  
VIDEO_LOCALITY_MODEL.md
```

5. assistant (2026-06-21T13:56:24.376Z)

The branch can't auto-pull (multiple branches issue), but that's fine ? I'll proceed read-only. Let me read all four docs.

6. user (2026-06-21T13:56:25.806Z)

```
1      # Internationalization (i18n) ? Kannada-first, by design  
2
```

```

3      **Status:** Design (P0 deliverable; phases P1+ are owner-gated, not started).
4      **Owner decision required** before any code lands ? see [Open decisions](#open-
decisions-for-the-owner).
5
6      > **? Implementation (2026-06-21):** this design is now being executed in the
**`khelsutra`** repo against
7      > today's surfaces ? tracked in **`khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`** ($7
P-tracker is the source of
8      > truth). P1's "static-UI retrofit" is **retargeted**: `backend/api/static` is gone, so
**P1 ? the vanilla
9      > `site/` (`t()` shim) and P3 ? the `web/` React SPA (react-i18next)**, both consuming
shared
10     > `locales/{en,kn,hi}` catalogs co-located in `khelsutra`. The framework-agnostic
foundation below is
11     > unchanged. Backend localization (P2) stays here in the engine repo, owner-gated.
12
13     ---
14
15     ## 1. Context ? why this, why now
16
17     The PMF beachhead is **Bangalore badminton academies**
([`PMF_MARKET_RESEARCH.md`](archives/field-pmf/PMF_MARKET_RESEARCH.md)).
18     Many academy coaches, parents, and players are far more comfortable in **Kannada** than
English.
19     A product that only speaks English adds friction exactly at the point of first contact.
So the
20     product must speak **Kannada first**, **Hindi next**, and extend to other Indian
languages
21     (Tamil, Telugu, Marathi, ?) **without a refactor**.
22
23     The owner's directive is precise: make the product **internationalizable _by design**.
Adding a
24     language must be "add a reviewed catalog" ? not "go re-touch the code." The
translations
25     themselves can wait; **the foundation that makes them cheap to add is the actual
deliverable.**
26
27     Today there is **zero i18n** in the repo (no i18n/locale/gettext/babel anywhere). Every
28     user-facing string is hardcoded English across four surfaces:
29
30     | Surface | Where | Examples (today) |
31     |---|---|---|
32     | **UI chrome** | `backend/api/static/{index.html,app.js,style.css}` (~40?50 strings) |
`"Validate & Index Video"`, `alert("Validation Failed! ?")`, `` `${count} Rally Segment${count}
> 1 ? 's' : ''} Selected` `` |
33     | **Backend coach-facing text** | `backend/main.py` `HTTPException(detail=?)`;
`backend/pipeline/validators/*` `message`; `backend/reporting/spec.yaml` `customer_verdict` +
per-rule `customer_message` | `Video file not found at '?'`; `Video quality rejected: Focus
check failed?`; `"? Your highlights are ready.` |
34     | **Self-navigating NL queries** (owner ask, future) | not built yet | "show me all
rallies > 15 shots where I won" ? the **query and its answer** must work in the user's language
|
35     | **Coach-mode recommendations** (owner ask, future) | not built yet | localized
recommendation text, not English |
36
37     A React/Vite SPA is planned to replace the static UI ([`UI_ALPHA_EXECUTION_PLAN.md`],
ADR-009)
38     in a **separate private repo `avidullu/khelsutra`**. The design below is therefore
deliberately
39     **framework-agnostic**: the catalog format and locale-negotiation contract are shared,
so the
40     static UI today and the SPA tomorrow consume the *same* translation source.
41
42     **Scope confirmed with owner:** UI + coach-facing backend text + (designed-for, built-
later)

```

```

43     NL/coach features. **Translations are Gemini-drafted, human-reviewed** ? this is
*inference*
44     use of a paid LLM, which the [licensing guardrail](#guardrail-alignment) explicitly
allows
45     (it is *not* a training-data source). The owner reviews the first draft of every string.
46
47     ---
48
49     ## 2. Core architecture
50
51     ### 2.1 The "by design" invariant ? no hardcoded user-facing strings
52
53     Every coach-facing string is a **stable key** resolved at render time against the active
locale.
54     Code references `t("compile.button")`, never the English prose. This is the single rule
that
55     makes "add a language = add a catalog" true, and it is **enforced by CI** (§2.6) so it
cannot
56     silently regress as the codebase grows.
57
58     ```
59     "Validate & Index Video"                ? t("video.action.validate_index")
60     alert("Validation Failed! ?review the checklist") ? t("toast.validation_failed")
61     `${count} Rally Segment${count>1?'s':''} Selected` ? t("compile.selected", {count}) #
ICU plural
62     HTTPException(detail="Video file not found at ?")? detail = tr(lang,
"error.video_not_found", {path})
63     ```
64
65     ### 2.2 One canonical English source catalog, per-locale siblings
66
67     ```
68     locales/
69     en/  common.json  # source of truth ? devs author keys here
70     kn/  common.json  # Kannada (Gemini-draft ? human-review)
71     hi/  common.json  # Hindi (next; first exercise of the extensibility checklist)
72     ```
73
74     - **Format: i18next-flavored JSON** ? works for the vanilla shim today *and*
react-i18next in
75     the SPA, with no conversion.
76     - **ICU MessageFormat** for interpolation and plurals: `{count, plural, one {# rally}
other {# rallies}}`.
77     This matters immediately ? the current UI already hand-rolls English plurals
78     (`Rally Segment${count > 1 ? 's' : ''}`), and Kannada/Hindi plural rules differ from
English.
79     ICU encodes each language's rules in the catalog, not the code.
80     - `en` is the source of truth; **every other locale is keyed identically** (enforced by
the
81     key-parity check in §2.6).
82
83     ### 2.3 Locale-negotiation contract (shared frontend ? backend)
84
85     One resolution order, defined **in one place** and honored by both UI and backend:
86
87     ```
88     persisted user preference ? UI switcher / ?lang=<code> ? Accept-Language header ?
default "en"
89     ```
90
91     - Supported locales are enumerated in **a single module** (one list to grow). Anything
outside
92     it falls back to `en`.
93     - The backend localizes its own responses (`HTTPException.detail`, report `customer_`)
from the

```

```

94     same `lang` query param / `Accept-Language` header, so a Kannada UI gets Kannada
errors.
95     - **Always-fallback to `en`** for any missing key ? a missing translation must degrade
to
96     English, never to a blank or a raw key in front of a user.
97
98     ### 2.4 Backend text ? keys + params, not prose
99
100    Validator `message`s and API errors today bake in English prose *and* interpolated
values
101    (`f"?sharpness score of {avg} is below ? threshold of {threshold}."`). The design
separates
102    the two:
103
104    - The validator/endpoint emits a **stable code + a params dict**
(`("blur.below_threshold",
105     {"score": avg, "threshold": threshold})).
106    - The **prose lives in the catalog**, looked up at the API boundary in the negotiated
locale.
107    - `ValidationResult` keeps an English `message` for logs/back-compat, but gains an
optional
108    `message_key` + `message_params`; the report/serializer prefers the key when present.
This
109    keeps internal/technical messages English (good for debugging) while customer-facing
text
110    localizes.
111
112    `spec.yaml` `customer_verdict` and per-rule `customer_message` become **keyed** (or per-
locale
113    overlays keyed by rule `code`); the technical `message` blocks stay English (they cite
114    `README#Qn` and are for the operator, not the end user).
115
116    ### 2.5 Indic font strategy ? (open decision ? see §4)
117
118    The bundled UI/watermark fonts (Outfit, Roboto Slab) **do not cover Kannada or
Devanagari**.
119    **Noto Sans Kannada / Noto Sans Devanagari are OFL-licensed**, which is the same tension
we hit
120    with the watermark font: OFL is not in the strict **MIT/Apache/BSD** code/data/weights
guardrail.
121    Fonts are *media assets*, not code or model weights, and OFL is explicitly designed for
122    unrestricted redistribution ? so the cleanest path is to **carve out an explicit OFL-
for-fonts
123    exception**, the same way we'd treat any bundled media. Options put to the owner in §4.
124
125    Note the **video watermark already takes a `font_path`** (`WatermarkConfig.font_path`,
126    `backend/pipeline/stitcher/compiler.py`), so stamping a Kannada/Hindi watermark is a
127    *configuration* change (point it at a Noto font) once the font decision is made ? no new
code.
128
129    ### 2.6 CI enforcement (the guardrail that keeps it "by design")
130
131    Two checks, both pure-Python and runnable in this repo (no GPU/torch needed):
132
133    1. **Key-parity test** ? every locale catalog has *exactly* the `en` key set; fail on
missing
134    keys, extra keys, or untranslated placeholder markers (e.g. a ` "__MISSING__" `
sentinel left
135    by the draft script). This is also the **extensibility proof**: drop a stub `ta.json`
with
136    one key and CI immediately lists every gap.
137    2. **No-bare-strings lint** ? a check that flags user-facing string literals not routed
through
138    `t()` / the backend `tr()` (scoped to the UI handlers and the API/validator/report
text

```

```

139     paths). Starts as a curated allowlist so it can land without boiling the ocean.
140
141     ### 2.7 Translation pipeline
142
143     ```
144     dev authors en keys ? scripts/i18n_draft.py (Gemini, INFERENCE only) ? kn/hi draft
145                        ? native-speaker review (owner reviews first draft)
146                        ? CI key-parity + placeholder check ? merge
147     ```
148
149     `scripts/i18n_draft.py` reads `locales/en/*.json`, asks Gemini for each missing key in
150     the
151     target locale with the key name and any ICU placeholders preserved, and writes a
152     draft
153     catalog with every machine-drafted value flagged for review. It never touches `en`,
154     never
155     invents keys, and is a *drafting aid* ? a human reviews before merge. (Reuses the
156     existing
157     Gemini provider credentials/config; this is the same "paid LLM = inference, never
158     training"
159     posture as the rest of the system.)
160
161     ---
162
163     ## 3. Phased execution (owner-gated; ONE PR per slice, off `master`)
164
165     | Phase | Scope | Repo | Gate |
166     |---|---|---|---|
167     | P0 | This design doc (`docs/I18N_PLAN.md`) | this | the immediate
168     deliverable |
169     | P1 | Foundation + static-UI retrofit: `locales/` + `en` source catalog
170     (extract the ~40?50 UI strings ? keys), a tiny vanilla t(key, vars) shim + `data-i18n`
171     wiring in `index.html`/`app.js`, language switcher (en/kn) + `localStorage` persistence +
172     `?lang=`, the font decision applied, and the kn catalog (Gemini-draft ? owner review).
173     Reuse the existing static mount / `serve_index` in `backend/main.py`. | this | owner approval |
174     | P2 | Backend + report localization: `backend/i18n/` with `tr(lang, key,
175     params)` (same catalog format), refactor validator `message`s to keys+params, localize
176     `HttpException.detail` + `spec.yaml` `customer_verdict`/`customer_message`, wire `Accept-
177     Language`/`?lang=`. | this | owner approval |
178     | P3 | SPA adoption: react-i18next consuming the same `locales/` catalogs;
179     documented FE?BE contract. | `avidullu/khelsutra` | needs repo access; or owner/Opus executes
180     there |
181     | P4 | Conversational + coach-mode (future): locale-aware NL query intake ?
182     templated localized answers; coach-mode recommendation templates. Ties to the owned-
183     model / PLATFORM $5A + rich-metadata work. | this + model | gated on those features existing |
184
185     Extensibility checklist (this is the by-design contract ? runs the same every
186     time):
187     > add `locales/<code>/common.json` ? `scripts/i18n_draft.py --locale <code>` ? human
188     review ?
189     > add <code> to the supported-locales list ? confirm font coverage ? CI key-parity
190     green.
191
192     Hindi (`hi`) is the first real exercise of this checklist ? if adding `hi` requires
193     touching
194     anything other than the steps above, the design failed and we fix the design, not paper
195     over it.
196
197     ---
198
199     ## 4. Open decisions (for the owner)
200
201     1. Target-UI scope ? confirm the shared framework-agnostic foundation (prove on
202     the static
203     UI now, SPA adopts the same catalogs later). Alternative: scope only to the SPA and

```

```

skip the
181     static-UI retrofit. _Recommendation: shared foundation ? it's the cheap, low-risk way
to
182     validate the catalog/negotiation contract before the SPA exists._
183 2. **Indic font licensing** ? (a) **allow OFL for bundled fonts** and self-host Noto
Sans
184     Kannada/Devanagari _(recommended; treats fonts as media, not code/weights); (b) rely
on
185     **system Indic fonts** (no bundle, inconsistent rendering across machines); (c)
**Google Fonts
186     CSS** (runtime external dependency, privacy/offline concerns for a local-first
product). This
187     also unblocks the **Kannada/Hindi video watermark** (needs a Noto `font_path`).
188 3. **v1 locale set & fallback** ? confirm `en` + `kn` for v1, `hi` next; always-fallback
to `en`
189     for missing keys.
190 4. **Backend-localization depth for v1** ? recommendation: localize the **customer
verdict /
191     report `customer_***` and **validation rejection reasons** first (highest user
impact);
192     defer technical/operator messages (they stay English by design).
193
194 ---
195
196 ## 5. Key files / seams
197
198 - **Localize:** `backend/api/static/{index.html,app.js,style.css}`; `backend/main.py`
199   (`HTTPException.detail` + the `serve_index`/static mount);
`backend/pipeline/validators/`
200   (`message` ? key+params; see `validators/base.py::ValidationResult`);
201   `backend/reporting/spec.yaml` (`customer_verdict` + per-rule `customer_message`).
202 - **New:** `locales/{en,kn,hi}/`, `backend/i18n/` (`tr()` + supported-locales +
negotiation),
203   `scripts/i18n_draft.py`, a CI key-parity test under `tests/` + a no-bare-strings check
wired
204   into `.github/workflows/ci.yml`.
205 - **Fonts:** reuse the bundled-asset pattern in `backend/assets/fonts/` (Apache-2.0
Roboto Slab
206   ships there today); add Noto Indic fonts there **iff** decision $4.2(a).
207 - **SPA:** `avidullu/khelsutra` (react-i18next) ? separate repo.
208
209 ---
210
211 ## 6. Verification
212
213 - **CI key-parity test** (Python, this repo): every locale has every `en` key; fail on
214   missing/extra/placeholder. Drop a stub `ta.json` ? CI flags every gap (the
extensibility proof).
215 - **Static UI:** load `?lang=kn` ? rendered strings resolve from the `kn` catalog;
switcher
216   persists across reload; any missing key falls back to English (no raw keys shown).
217 - **Backend:** request with `Accept-Language: kn` (or `?lang=kn`) ? localized error
`detail` +
218   report `customer_`; unit tests for fallback, ICU interpolation, and plural rules.
219 - **Fonts:** Kannada + Devanagari render correctly ? **manual eyeball on the owner's
machine**
220   (the cloud container can't render fonts).
221 - **Docs-only P0 PR:** internal links resolve; indexed in [`docs/README.md`](README.md);
222   cross-linked from [`PLATFORM_ARCHITECTURE.md`](PLATFORM_ARCHITECTURE.md) (the NL/coach
surfaces)
223   and [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) (the Bangalore beachhead).
224
225 ---
226
227 ## Guardrail alignment

```

```

228
229 - **Paid LLM = inference only.** Gemini drafts translations (inference); it is **never**
a
230 training-data source. ? consistent with [`.CLAUDE.md`](../CLAUDE.md) and the ratified
guardrail.
231 - **Licensing.** Code stays MIT/Apache-2.0/BSD. The *only* tension is **fonts** (OFL) ?
handled
232 as an explicit, owner-approved media-asset exception in §4.2, not a silent dependency.
233 - **No minors / biometrics scope creep.** i18n touches presentation only; it introduces
no new
234 personal-data collection.
235
236 [`.UI_ALPHA_EXECUTION_PLAN.md`]: ../UI_ALPHA_EXECUTION_PLAN.md
237

```

7. user (2026-06-21T13:56:25.955Z)

```

1      # Friendly & Multilingual ? non-tech-savvy UX + Hindi/Kannada localization
2
3      > **Status:** `DRAFT` ? owner-gated (implementation starts only on explicit approval) .
**Owner:** Avi Dullu . **Created:** 2026-06-21 . **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE).
5      > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; thread in `NEXT_STEPS.md`; pointer in engine `docs/DOC_STATUS.md`; memory
`current-state` + `next-session-ux-localization`.
6      > **Relation to existing docs:** this is the **implementation project** for two engine
design docs ? it *adapts & supersedes-in-practice* `badminton-highlight-
indexer/docs/I18N_PLAN.md` (localization) and the UX half of `?/docs/UI_PROPOSAL.md` (non-tech-
savvy console). Those stay the canonical *design*; this doc carries the *tracked execution*
against today's surfaces.
7      > **Honesty note:** each claim tagged `[verified]` (checked against code this session),
`[researched]` (needs sign-off), or `[design]` (proposed). Not legal advice ? the OFL font
question (D-FONT) is owner/counsel-gated.
8
9      ---
10
11     ## 0. TL;DR
12     A coach or parent in Bangalore should land on KhelSutra, feel at home, set up their
camera with confidence, and get highlights ? **in Kannada or Hindi**, with **zero hand-
holding**. This project isolates that goal into one tracked workstream-pair: ** (A) ** make the UI
super friendly for non-tech-savvy users, and ** (B) ** localize to Hindi + Kannada. Both surfaces
(`web/` React SPA + `site/` vanilla marketing) have **zero i18n today** `[verified]`, so the
foundation ? shared `{en,kn,hi}` catalogs + a negotiation contract + CI parity ? is the
real deliverable; translations follow. **The single most important caveat:** translations are
**human-reviewed** (Gemini drafts are *inference only, never training* ? the standing
guardrail), and Kannada/Hindi rendering is **blocked on the OFL font decision (D-FONT)**. The
cheapest unblocked high-impact first slice is the **SPA i18n scaffold + language switcher (en-
only)** ? it locks the shared catalog/negotiation contract and adds the render-test safety net
the SPA currently lacks.
13
14     ## 1. Problem & goal
15     **Why now.** The public site (kheelsutra.guru) is live, clean, honest, and tested. The
PMF beachhead is **Bangalore badminton academies**; many coaches, parents, and players are far
more comfortable in **Kannada** than English, and most are **not highly tech-savvy** `[verified
? design intent]` (`UI_PROPOSAL.md:25,28`, `I18N_PLAN.md:10-13`, `PMF_MARKET_RESEARCH.md:116`,
`KHELSUTRA_PRODUCT_OVERVIEW.md:13/50/61`). An English-only, knob-heavy product adds friction at
the exact point of first contact.
16
17     **Concrete outcome ("good" looks like):** a Kannada/Hindi-comfortable, phone-first, non-
technical coach/parent can ? without help ? (1) understand what the product does and how to
record reliably, (2) request access / use the console, (3) pick rallies and get a reel, and (4)
share it on WhatsApp, **entirely in their language**.
18
19     **Read to get current:** this doc's §3§4; the engine design `I18N_PLAN.md` (canonical

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```

i18n design) + `UI_PROPOSAL.md` $2 (personas, the 6 alpha screens, "teach recording *before*
upload"); `VIDEO_RECORDING_GUIDELINES.md`; this repo's `PER_RALLY_SELECTION.md` (the honest SPA
scope) + `REQUEST_ACCESS_FORM.md`.
20
21     ## 2. Decisions locked (structural ? owner directive 2026-06-21)
22     | # | Decision | Source / date | Implication |
23     |---|-----|-----|-----|
24     | D1 | **Isolate as ONE tracked project** spanning both workstreams (A: UX, B: kn/hi
l10n) | owner, 2026-06-21 | This doc; $7 = source of truth; archived on DoD |
25     | D2 | **Home = `khelsutra/docs/`** (the implementation repo) | owner-gated tracked-doc
convention | `web/` + `site/` both live here; $7 tracks khelsutra PRs; engine P2 backend-l10n is
a cross-repo deferred row |
26     | D3 | **Adapt `I18N_PLAN.md` to today's surfaces:** P1 (vanilla `t()` shim) ? `site/`;
P3 (react-i18next) ? `web/`; **drop the dead `backend/api/static` retrofit** | `[verified]` the
doc targets `backend/api/static` (`I18N_PLAN.md:25,30,156,191`) which exists only in the engine
repo, not here | The doc's framework-agnostic *foundation* (catalog format + negotiation + ICU
plurals + CI) survives unchanged; only the wiring re-points |
27     | D4 | **Engine P2 backend-localization stays owner-gated + cross-repo** (not in this
DoD) | `CLAUDE.md` guardrails | Dynamic engine error/report text localizes later; tracked as a
deferred row |
28     | D5 | **Archive trigger = en+kn+hi live on BOTH surfaces** (+ the non-tech UX
baseline), then run the engine `DOC_STATUS.md` $2` archival checklist | owner, 2026-06-21 | See
$9 |
29     | D6 | **D-LOCALES: `en` + `kn` for v1, `hi` next; always-fallback to `en`** | owner,
2026-06-21 | Kannada-first beachhead; `hi` = catalog-only (both 2-form CLDR) |
30     | D7 | **D-FONT: OFL-for-media carve-out ? self-host Noto Sans Kannada/Devanagari**
(counsel may still confirm) | owner, 2026-06-21 | Unblocks B4/B5/A3 + the kn/hi reel watermark;
an explicit exception to the MIT/Apache/BSD code guardrail (fonts = media) |
31     | D8 | **D-SURFACE-FIRST: lead with the `web/` SPA scaffold; SKIP the dead static-UI
retrofit** | owner, 2026-06-21 (via B1 go-ahead) | B1 is greenlit; site recording-education
localized early |
32     | D9 | **D-CATALOG-HOME: co-locate `locales/` + CI in `khelsutra`** (shared, consumed by
SPA + site) | owner, 2026-06-21 (via B1 go-ahead) | No cross-repo coupling |
33
34     *Only **D-BACKEND-DEPTH** ($8.3) remains open ? and it's a deferred, cross-repo engine
concern (B7), not a v1 blocker.*
35
36     ## 3. Foundation ? research / prior art
37     **Current-state facts `[verified]` this session** (adversarially checked against the
working tree):
38     - **`web/` SPA: zero i18n.** `web/package.json` deps are only `react`/`react-dom`; no
`i18next`/`react-intl`/`formatjs`; no `locales/` dir; no `t()`/`useTranslation`/`data-i18n`
anywhere. User-facing English is hardcoded in `.tsx` (e.g. `QueryBar.tsx:46-47,56,72,76,78`;
`App.tsx:115,211,224`) ? ~50 literals + ~16 derived labels.
39     - **`site/` marketing: vanilla, zero i18n.** Three `first, Hindi-next ? not Kannada-only.
42
43     **Localization prior art `[researched]`:**
44     - **CLDR plural categories:** Hindi = `{one, other}`; Kannada = `{one, other}`. Both are
2-form (same shape as English) ? once ICU MessageFormat is in the catalog, adding `hi`/`kn`
plurals is **catalog-only**, no code. (ICU still required because English `count>1`'s`:` hand-
rolling ? present today in the SPA ? doesn't generalize.)
45     - **react-i18next** (web/): `i18next` + `react-i18next` + `i18next-browser-
languagedetector` + an ICU/plural post-processor; lazy-load per-locale JSON; persist via
`localStorage`.

```



```

46 - **Shared catalog + tiny shim** (site/): the SAME i18next-flavored JSON consumed by a
~30-line `t(key, vars)` shim that walks `data-i18n` attributes ? no build step, no framework,
mirrors i18next interpolation/plural rules for the 2-form languages.
47 - **Noto Sans Kannada / Noto Sans Devanagari** are **OFL-1.1**; OFL permits commercial
bundling/redistribution (fonts = media, not code/weights). Self-host (e.g. `@fontsource`) with
`font-display: swap` + subsetting. **Gated on D-FONT.**
48 - **Non-tech-savvy / HCI4D patterns:** icon **+ text** (never icon-only), non-color
selection indicators, large touch targets, mobile-first, WhatsApp as the share channel, a
prominent persistent language switcher.
49
50 ## 4. Design / architecture
51 **Two surfaces, one shared source of truth.**
52
53 ```
54     locales/                                # shared catalogs ? co-located in khelsutra (D-CATALOG-
HOME, see §8)
55     en/   common.json                      # source of truth ? devs author keys here
56     kn/   common.json                      # Kannada   (Gemini-draft ? human-review)
57     hi/   common.json                      # Hindi     (next ? the extensibility proof)
58
59     web/   ? react-i18next                 # I18N_PLAN P3: provider in main.tsx; useTranslation in
components
60     site/  ? t(key,vars) shim              # I18N_PLAN P1 retargeted: data-i18n attrs over the
static HTML
61     ```
62
63 - **Negotiation contract (one order, both surfaces):** `persisted pref (localStorage) ?
switcher / ?lang=<code> ? navigator/Accept-Language ? "en"`. Supported locales in a single
module; anything else falls back to `en`; **a missing key degrades to English, never to a raw
key.**
64 - **Keys, never prose** ? `t("compile.button")`, with ICU for interpolation/plurals:
`"{count, plural, one {# rally} other {# rallies}}"` . This replaces the SPA's current hand-
rolled `Segment${count>1?'s':''}`.
65 - **Translation pipeline (guardrail-clean):** dev authors `en` keys ?
`scripts/i18n_draft.py`-style Gemini **draft** (inference only, never training; flags every
value for review) ? **owner/native-speaker review** ? CI key-parity ? merge. Gemini never
touches `en`, never invents keys.
66 - **CI enforcement (keeps it "by design")** (1) **key-parity** ? every locale has
exactly the `en` key set (drop a stub `ta.json` ? CI lists every gap = the extensibility proof);
(2) **no-bare-strings** ? flags user-facing literals not routed through `t()` (curated allowlist
to start).
67 - **Fonts:** bundle Noto Sans Kannada + Devanagari **iff D-FONT = OFL carve-out**; this
also unblocks a Kannada/Hindi **reel watermark** (engine `WatermarkConfig.font_path` already
takes a path ? config, not code).
68
69 **Non-tech-savvy UX levers** (ranked impact × effort, grounded in personas; surface-
tagged):
70
71 | Lever | Surface | Impact | Effort | Note |
72 |---|---|---|---|---|
73 | Plain-language copy ? strip engine jargon, keep honest caveats | both | high | small | banners + footer carry backend vocabulary today |
74 | Localized **WhatsApp share** for the downloaded reel | both | high | small | WhatsApp is **the** channel for this audience |
75 | **Language switcher** persisted via a shared localStorage key | both | high | small | site promises kn/hi but ships no switcher |
76 | Localize **recording-education** (capture + gist) kn?hi | site | high | medium | gates input quality; the highest-leverage kn content |
77 | Mobile-first touch targets; tooltip-text ? visible | web | high | medium | tiny targets + desktop layout + invisible tooltips on phones |
78 | Icon **+ text** + non-color selection indicator | web | medium | small | never icon-only |
79 | Friendly engine-error mapping over raw FastAPI `detail` | web | medium | medium | `CompileResult` passes engine text untranslated |
80 | Plain-language framing for the Request-Access GPU/storage Qs | site | medium | small |

```

```

"GPU"/"S3/GCS" are intimidating |
81 | Background-processing reassurance + status as icon chips | web | medium | small |
honest expectations on slow uplinks |
82 | Fix English-source recording gaps (a diagram) before translating | site | medium |
medium | the diagram is reusable across all 3 languages |
83 | Tap-only filter controls alongside the English query bar | web | medium | large | the
`parseQuery()` parser is English-grammar only |
84 | Keep the honest *today-vs-roadmap* split through translation | both | medium | small |
never soften "experimental"/"not available yet" |
85
86 **Reuse map (what's new vs reused):** *Reused* ? the i18next-flavored JSON + negotiation
contract + ICU plurals + key-parity/no-bare-strings CI + draft?review pipeline (all from
`I18N_PLAN.md`); the shipped recording pages; the SPA's `lib/format.ts`/`lib/query.ts`;
`site/src/assets.test.js` + the SPA's vitest harness. *New* ? `locales` catalogs; the SPA
react-i18next provider + switcher; the site `t()` shim + `data-i18n` wiring; Noto font bundle
(gated); a jsdom render-test env for the SPA.
87
88 ## 5. Privacy / scope note
89 i18n touches **presentation only** ? no new personal-data collection, no biometrics,
minors-excluded posture unchanged. WhatsApp share opens the user's own client with a pre-filled
message; it sends nothing server-side. Gemini-draft translation is inference, not training (no
user data in the loop).
90
91 ## 6. Honest limits ? what this does NOT do
92 - A green CI / DONE status proves **key-parity + render + no-bare-strings**, **not**
translation *quality* ? that's human-gated and owner-paced.
93 - Kannada/Hindi will **not render** until the Noto fonts land (D-FONT); until then kn/hi
catalogs exist but display tofu/fallback.
94 - The SPA is **job-tethered** ; localizing **dynamic engine errors/report text** needs
engine P2 (owner-gated, cross-repo, D-BACKEND-DEPTH) ? out of this DoD.
95 - The query bar stays **English-grammar parsing** ; kn/hi users rely on tap-only controls
(a later, larger slice), not free-text Kannada queries.
96 - **Mobile-perfect** layout is an alpha non-goal (usable-on-phone is in scope; pixel-
perfect is not).
97
98 ## 6.5 Testing strategy ? automatic language negotiation (rigorous, owner-required)
99 **The bar (owner, #41):** a user whose **computer/browser language is Kannada or Hindi
must get the respective version automatically** ? that is the core promise and it is **proven by
tests at every layer, never assumed.** Auto-selection IS the negotiation order (persisted pref ?
`?lang` ? browser/`Accept-Language` ? `en`); each step gets a test.
100
101 **Negotiation matrix ? every row is an automated test:**
102
103 | # | Signal under test | Setup | Expected | Proven in |
104 |---|---|---|---|---|
105 | N1 | **Browser/OS = Kannada** | no persisted pref, no `?lang`,
`navigator.languages=["kn-IN", ?]` | resolves to **kn** ? Kannada content (once the kn catalog
ships) | **B1** detection . **B5** content |
106 | N2 | **Browser/OS = Hindi** | `navigator.languages=["hi-IN", ?]` | resolves to **hi**
? Hindi content | **B6** (hi added to the supported set + catalog) |
107 | N3 | **`?lang` override** | `?lang=kn`, browser = en | kn | B1 |
108 | N4 | **Persisted pref wins** | `localStorage=kn`, browser = en, no `?lang` | kn (a
returning user keeps their choice) | B1 |
109 | N5 | **Region subtag normalizes** | `navigator.language="kn-IN"` | base **kn** (not a
"kn-IN" miss) | B1 |
110 | N6 | **Unsupported language** | `navigator="ta-IN"` (Tamil, not yet supported) |
**en** fallback ? never a locale we can't serve | B1 |
111 | N7 | **Missing key in active locale** | kn active, a key absent from `kn/common.json`
| **en** fallback for that key (no raw key shown) | B1/B5 |
112 | N8 | **Static site (vanilla shim)** | `Accept-Language: kn` / `?lang=kn` on a `site/`
page | kn-rendered page (same order as the SPA) | B3 |
113 | N9 | **Backend (engine P2)** | `Accept-Language: kn` to the API | kn customer/report
text | B7 (owner-gated) |
114
115 **Levels & tooling:**

```

116 - ****SPA** (`web/`):** jsdom tests stub `navigator.language`/`navigator.languages`,
`localStorage`, and the `?lang` query ? assert `i18n.language`/`resolvedLanguage` + rendered
text per the matrix. *(B1 lands N1/N3?N7 at the ****detection**** level now ? `kn` is supported; the
kn/hi ****content**** assertions ride the catalogs in B5/B6.)*

117 - ****Static site** (B3):** the shared `t()` shim's negotiation is unit-tested the same way;
plus a server test that `Accept-Language: kn`/`?lang=kn` returns the kn-rendered page.

118 - ****CI key-parity****:** guarantees every locale has every key, so a correctly-negotiated
locale can never fall through to a raw key.

119 - ****End-to-end** (manual, owner machine ? per I18N_PLAN \$6):** set the real OS/browser to
Kannada, then Hindi ? load both surfaces ? confirm the right language renders ****and** the Noto
fonts display** (the cloud CI container can't render fonts; the owner eyeball is the final
gate).

120 - ****? Query parsing / semantic understanding** (forward-looking, owner directive
2026-06-21) ? i18n testing is **MANDATORY** there.** The query bar is ****English-grammar-only**** today
(a documented limit, tracker A9): the user types English queries (`over 10s`, `top 15 longest`)
and a localized ****sample-query prompt**** shows them what to type in ***every*** locale (kn/hi
included). The ****moment**** the parser gains ****semantic understanding**** or accepts ****Kannada/Hindi**
query input** (e.g. typing a Kannada/Hindi phrase, or a conversational/NL builder ? I18N_PLAN
P4), i18n tests become a ****merge gate****: a kn/hi query (typed or spoken) must parse to the
****correct selection****, the localized ****parse-feedback**** ("parsed-pill") must render in the
active locale, and the unsupported/empty-result messages must localize. ****No query-NL /**
semantic-parsing work merges without those tests.**

121

122 A locale that's reachable by the switcher but ****not auto-selected** from the browser
language is a bug, not a "later".** This matrix is a DoD gate (\$9).

123

124 **## 7. Deliverables & progress tracker** ? ****source of truth****

125 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. ****One small PR per row****,
branched from `origin/master`, no stacking.

126

ID	Deliverable	Workstream	Depends on	Gated?	Status	PR
P0	This tracked project doc + inbound pointers				No	
[#41](https://github.com/avidullu/khelsutra/pull/41)	engine					
[#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275)						
B1	**SPA i18n scaffold + language switcher, en-only** (react-i18next + detector + `en` catalog + switcher + persistence + **jsdom render tests** + **browser-language negotiation tests** \$6.5 N1/N3?N7)	B	P0	No		
[#42](https://github.com/avidullu/khelsutra/pull/42)						
B2	Extract SPA strings ? `en` catalog (~50 literals + ~16 derived; ICU plural keys); add **key-parity + no-bare-strings CI**	B	B1	No		
[#45](https://github.com/avidullu/khelsutra/pull/45)						
A1	Plain-language copy pass on the `en` catalog (de-jargon; icon+text; non-color selection)	A	B2	No		
B3	Tiny shared `t()` shim + `data-i18n` plumbing for `site/`, en-only (proves shared catalogs across surfaces)	B	B2	D-CATALOG-HOME		
A2	Localized **WhatsApp share** for the reel (SPA + site)	A	B1/B3	No		
A6	Plain-language framing for Request-Access GPU/storage questions	A	B3	No		
A7	Fix English-source recording gaps (add a placement diagram) before localizing	A				
B4	**Noto Sans Kannada + Devanagari** bundle (OFL carve-out) ? *SPA done; reel watermark `font_path` = engine follow-up	B		D-FONT		
[#46](https://github.com/avidullu/khelsutra/pull/46)						
B5	**Kannada (`kn`) catalog** ? render ? *SPA done (AI-draft, native review pending); `site/` rides B3*	B	B2/B4	D-LOCALES		
[#47](https://github.com/avidullu/khelsutra/pull/47)						
A3	Localize **recording-education** (capture + homepage gist) to kn	A	B3/B5	D-LOCALES+D-FONT		
A4	Mobile-first touch targets + tooltip-text ? visible (SPA)	A	B1	No		
A5	Friendly engine-error mapping over raw FastAPI `detail` (SPA, client-side map)	A	B2	No		
B6	**Hindi (`hi`) catalog** ? render (the extensibility proof ? just a catalog) ? *SPA done (AI-draft, native review pending)*	B	B5	D-LOCALES		
[#48](https://github.com/avidullu/khelsutra/pull/48)						

143 | A8 | Background-processing reassurance + verdict/status as icon chips (SPA) | A | B2 |
No | ? | ? |

144 | A9 | Tap-only filter controls alongside the English query bar (SPA) | A | B2 | No | ?
| ? |

145 | B7 | *(deferred, cross-repo)* Engine **P2 backend customer-facing localization**
(report `customer\_` + rejection reasons) | B | engine | **D-BACKEND-DEPTH, owner-gated** | ? |
? |

146

147 *** MILESTONE ? SPA fully trilingual (2026-06-21):** the `web/` Studio SPA now speaks
en + kn + hi end to end (B1?B2?B4?B5?B6, all merged). A **Kannada-default browser auto-
renders Kannada**, and the switcher walks kn ? hi ? en ? **proven by a Playwright screencast**
with `locale: kn-IN` (`kheldutra-i18n-screencast.mp4`; frames captured per locale + `
lang>` tracked). This meets the owner's success criterion **on the SPA surface**. **Still open
for the full §9 DoD:** **B3** (the `site/` marketing surface via the shared `t()` shim ? the
homepage/capture pages are still English-only), **A3** (recording-education in kn/hi), the
non-tech UX A-items, and a **native-speaker review** of the AI-drafted kn/hi catalogs before
a wide public launch. Do **not** archive until both surfaces are trilingual.

148

149 ## 8. Open questions ? owner (blocking gates) `[design]`

150 *Status (2026-06-21): 4 of 5 **LOCKED/ADOPTED** by the owner ? see §2 (D6?D9). Only
D-BACKEND-DEPTH remains open (and it's deferred, not a v1 blocker).*

151 1. **D-FONT ? ? LOCKED ? (a) OFL-for-media carve-out, self-host Noto Sans
Kannada/Devanagari** (counsel may still confirm). An explicit exception to the MIT/Apache/BSD
code guardrail (fonts = media). Unblocks B4/B5/A3 + the kn/hi reel watermark.

152 2. **D-LOCALES ? ? LOCKED ? `en` + `kn` for v1, `hi` next, always-fallback to `en`**
(both 2-form CLDR ? `hi` is catalog-only).

153 3. **D-BACKEND-DEPTH ? ? OPEN (deferred, not a v1 blocker).** Recommend **customer
verdict/report `customer\_` + validation-rejection reasons first; technical/operator text stays
English; phase two** (engine repo, owner-gated ? tracker row B7).

154 4. **D-SURFACE-FIRST ? ? ADOPTED ? lead with the SPA scaffold (B1), localize site
recording-education early, SKIP the dead static-UI retrofit.**

155 5. **D-CATALOG-HOME ? ? ADOPTED ? co-locate `locales/` + CI in `kheldutra`** (a shared
i18n dir consumed by both the SPA and the site). (The engine `I18N\_PLAN` assumed the indexer
repo; this is the adaptation.)

156

157 ## 9. Definition of done

158 - [] **Automatic language negotiation (§6.5) is green** ? a **Kannada browser gets
Kannada and a Hindi browser gets Hindi** (N1/N2); persisted-pref / `?lang` / region-subtag /
unsupported-fallback / missing-key-fallback all proven (N3?N7) on the SPA, and the static-site
shim honors `Accept-Language` + `?lang` (N8).

159 - [] **Manual cross-check (owner machine):** OS/browser set to Kannada and to Hindi
each render the respective language on BOTH surfaces, with Noto fonts displaying.

160 - [] `en + kn + hi` catalogs live and **rendering** on BOTH `web/` SPA and `site/`
marketing (switcher + persistence + `en` fallback; no raw keys shown).

161 - [] **Noto Sans Kannada + Devanagari render correctly** ? owner eyeball on a real
machine.

162 - [] CI **key-parity + no-bare-strings** green; a stub locale flags every gap.

163 - [] **Recording-education** (capture + homepage gist) available in `kn` + `hi`.

164 - [] **Non-tech UX baseline:** plain-language copy (no raw engine jargon in user-facing
strings), mobile-usable touch targets, icon+text, WhatsApp share, honest today-vs-roadmap
preserved through translation.

165 - [] §7 all rows ? (or explicitly deferred with a reason) ? flip **Status ? DONE** +
completion note ? run engine `DOC\_STATUS.md §2` archival checklist ? `git mv` to
`docs/archives/`.

166

167 ## 10. References

168 **Internal code anchors `[verified]`:** `web/package.json`,
`web/src/{App.tsx,main.tsx,components/QueryBar.tsx,components/CompileResult.tsx,lib/query.ts}`;
`site/public/{index,capture,own-model}.html`, `site/server.js`, `site/src/assets.test.js`.
This repo's docs: `docs/PER\_RALLY\_SELECTION.md`, `docs/REQUEST\_ACCESS\_FORM.md`,
`docs/README.md`, `NEXT\_STEPS.md`. **Engine design docs (canonical):** `badminton-highlight-inde
xer/docs/{I18N_PLAN.md,UI_PROPOSAL.md,VIDEO_RECORDING_GUIDELINES.md,PMF_MARKET_RESEARCH.md,PMF_F
IELD_SIGNALS.md,KHELSUTRA_PRODUCT_OVERVIEW.md,DOC_STATUS.md}`. \*\*External `[researched]`:**
Unicode CLDR plural rules (kn/hi = one/other); react-i18next + i18next-icu; SIL Open Font
License 1.1; Google Noto Sans Kannada/Devanagari.

```

169
170 ---
171
172 ### Changelog
173 - `2026-06-21` ? created. Isolated the UX + Hindi/Kannada work into one tracked project
(owner directive). Grounded in a multi-agent research+verification pass (web/+site/ string maps,
Indic-l10n prior art, 5 adversarial verdicts). Decisions D1?D5 locked; 5 product/licensing
decisions surfaced as open; $7 tracker seeded with B1 as the recommended first slice.
174 - `2026-06-21` ? owner locked **D6 (en+kn?hi)** + **D7 (OFL Noto carve-out)** and
adopted **D8 (SPA-first)** + **D9 (catalogs in khelsutra)**; only D-BACKEND-DEPTH remains open.
**B1 greenlit ? ? in progress.**
175 - `2026-06-21` ? **B1 opened ? [#42](https://github.com/avidullu/khelsutra/pull/42)**:
SPA i18n scaffold (react-i18next + ICU + detector) + `en` catalog + language switcher + the
SPA's first jsdom render tests. Verified typecheck/79-tests/build green + live in-browser
(switcher persists, en-fallback). Doc/pointer PRs opened: khelsutra
[#41](https://github.com/avidullu/khelsutra/pull/41), engine
[#275](https://github.com/khelsutra/badminton-highlight-indexer/pull/275).
176 - `2026-06-21` ? **$6.5 Testing strategy added (owner ask, #41):** a rigorous
**automatic-negotiation matrix** (N1?N9) so a Kannada/Hindi **browser** auto-gets the respective
version ? proven at the detection level now (B1), at the content level with the catalogs
(B5/B6), on the static shim (B3), and by a manual font-render eyeball. Wired into the DoD ($9).
B1 (#42) gains the detection tests (N1/N3?N7).
177 - `2026-06-21` ? **? SPA MILESTONE: fully trilingual (owner directive "take it to
completion + screencast").** B2 (#45, full string extraction + key-parity/no-bare-strings CI) ?
B4 (#46, self-host OFL Noto) ? B5 (#47, Kannada) ? B6 (#48, Hindi ? the extensibility proof) all
merged. 99 web tests green incl. N1/N2 browser-detection + B5/B6 render + key-parity.
**Playwright screencast** (`locale: kn-IN`): a Kannada-default browser auto-renders Kannada ?
switch hi ? switch en, each with `<html lang>` tracked (`khelsutra-i18n-screencast.mp4` + per-
locale frames). kn/hi are **AI-drafted ? native review pending** . **Remaining for full $9 DoD:**
B3 (the `site/` marketing surface), A3 + the UX A-items, native-speaker translation review.
178

```

8. user (2026-06-21T13:56:26.205Z)

```

1      # UI Proposal ? khelsutra.guru web app (alpha ? beta)
2
3      **Status:** PROPOSAL ? awaiting owner ratification (no implementation starts without
explicit approval, per CLAUDE.md).
4      **Date:** 2026-06-10 . **Inputs:** PMF_MARKET_RESEARCH, COMMERCIALIZATION,
PLATFORM_ARCHITECTURE, DEFERRED_HARDENING_PLAN (#16), ROADMAP, VIDEO_RECORDING_GUIDELINES,
reporting/spec.yaml, current API + static UI.
5
6      > **? Partly shipped + in implementation (2026-06-21):** the alpha SPA now lives in
**`khelsutra/web/`**
7      > (the per-rally console shipped) and the marketing site in `khelsutra/site/`. The
**non-tech-savvy UX** half of
8      > this proposal is tracked in **`khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`**
(friendly + Hindi/Kannada). Treat
9      > the screens below as design intent; the live state is in that repo. *(Resolves
`DOC_STATUS.md` finding #5 for
10     > this doc; the static-UI references are historical ? that UI is retired.)*
11     **Companion:** `docs/HOSTING_PLAN.md` (domain, topology, costs).
12
13     ## 1. Strategic frame ? why a UI now, and what it secretly is
14
15     The UI is not just product surface; it is the **corpus-growth engine**. The single
highest-leverage
16     item in NEXT_STEPS is growing the golden corpus, and the v1 scope already commits to a
17     **human correction loop**. Every tester who accepts/rejects/nudges a rally boundary in
the review
18     screen produces exactly the labels the Gen-0/M0.x track is starved for (subject to the
per-user
19     training opt-in guardrail). 10?25 alpha testers ? a steady stream of labeled, real-
world,
20     multi-court-biased footage ? the distribution where the owned model's 0.66 ship target

```

```

lives.
21
22     Serving economics per the 2026-06-10 quality table: alpha serves the **Gemini path**
23     (0.93 on clean footage, ~$0.05/run) ? no GPU required in the request path. The
WASB/owned-model
24     path stays on the owner's box and is *not* exposed in the alpha UI (segmenter choice
hidden).
25
26     **The CLI is not replaced.** The web app talks to the same FastAPI surface the CLI
drives; the CLI
27     remains the power-user/ops tool.
28
29     ## 2. Personas (from PMF research)
30
31     - **Coach / academy (primary, B2B, India):** batches of match recordings, wants rally
review,
32     per-player reels for students/parents, per-rally stats. Already records on static
cameras.
33     - **Enthusiast (B2C funnel):** records own matches, wants a highlight reel + basic
stats, shares
34     with friends. Tripod discipline is the known adoption risk ? the UI must teach
recording
35     guidelines *before* upload, not fail videos after.
36
37     ## 3. ALPHA ? "quick & dirty" (goal: 10?25 invited testers, end-to-end, zero hand-
holding)
38
39     **Stack:** React + Vite + Tailwind SPA, deployed per HOSTING_PLAN (static hosting, $0).
40     API = existing FastAPI on the GPU box via tunnel at `api.khelsutra.guru`.
41     **Auth:** Cloudflare Access email-allowlist (OTP), free ?50 users, **zero auth code** ?
backend
42     reads the authenticated-email header for per-user scoping (alpha-grade tenancy: header
is
43     trustworthy because the tunnel only accepts Access-authenticated traffic).
44
45     ### Alpha screens (6)
46
47     | # | Screen | What it does | Backed by |
48     |---|-----|-----|-----|
49     | A1 | **Matches** (home) | User's videos, status chip per video (verdict ?/??/?),
upload CTA | `videos` + owner scoping (B4) |
50     | A2 | **Upload & Process** | Drag-drop; inline recording-guidelines checklist (pre-
upload education from VIDEO_RECORDING_GUIDELINES); chunked upload; then auto-starts processing |
B2 upload endpoint ? B1 job |
51     | A3 | **Job progress** | Stage list (validation ? segmentation ? AI confirm ? report)
with live status; verdict banner on completion | B1 `GET /api/jobs/{id}` poll |
52     | A4 | **Rally review** ? | Video player + rally list; click rally ? seek & play that
span; per-rally **Accept / Reject / nudge start?end ±0.5 s**; bulk "all good"; chips: duration,
shot count (+confidence) | B6 correction endpoints ? `candidate_segments.human_label/notes` ?
**the corpus engine** |
53     | A5 | **Highlights** | Select accepted rallies ? name reel ? compile ? download; list
of past reels | `/api/compile` + B3 download endpoint |
54     | A6 | **Report** | Renders the customer-audience report (verdict, notes, FAQ refs);
"nerd view" toggle shows technical.md | existing reporting artifacts |
55
56     ### Backend work the alpha actually requires (honest dependency list)
57
58     | ID | Item | Notes |
59     |----|-----|-----|
60     | B1 | **Async job model ? DEFERRED_HARDENING_PLAN #16** | The committed next PLATFORM
slice; `POST /api/process` ? 202 + `job_id`, `jobs` table (user_version 1?2),
ThreadPoolExecutor, `GET /api/jobs/{id}`. **Approving this proposal = greenlighting #16.** A
5?60 min blocking HTTP call cannot sit behind a tunnel UI. |
61     | B2 | **Chunked upload endpoint** | `POST /api/upload` accepting ?90 MB parts (free-
tier proxy body limit is 100 MB), assemble ? MD5 ? register under `allowed_video_root/<user>/`.

```

Size cap (e.g. 4 GB) + mp4/mov allowlist. |

62 | B3 | ****Download endpoint**** | `GET /api/highlights/{video\_id}/{name}` streaming the
compiled reel (today `/api/compile` returns a server-local path the browser can't reach). |

63 | B4 | ****Per-user scoping**** | `owner\_email` column on `videos` (migration), read from
the Access header, filter every query. Alpha-grade, single DB. |

64 | B5 | ****Retire/replace static UI**** | The SPA supersedes `backend/api/static` (which has
a known `rallies` vs `play\_segments` field bug ? don't fix, replace). |

65 | B6 | ****Correction-loop endpoints**** | `PATCH /api/segments/{id}` ?
accept/reject/boundary-nudge persisted to `candidate\_segments.human\_label`/`notes` (+
provenance: who, when). This is v1-scope "correction loop" made real. |

66 | B7 | ****Exposure hardening**** | CORS narrowed to the app origin; basic rate limit;
upload quota per user; `allowed\_video\_root` set; Gemini key rotated (already in NEXT_STEPS
housekeeping). |

67

68 **### Alpha non-goals (explicit)**

69

70 No payments, no orgs/teams, no athlete profiles, no share links, no owned-model serving,
no

71 multi-sport selector (badminton only), no notifications, no mobile-perfect layouts
(usable on

72 phone, optimized for laptop), no conversational QA (deferred in roadmap).

73

74 **### Alpha effort estimate**

75

76 B1 ? 2?3 sessions · B2?B7 ? 2?3 sessions · SPA (6 screens) ? 3?4 sessions ? ****~8?10**
working

77 sessions** to first tester invite. Sequencing: B1 ? B2/B3/B4 ? SPA skeleton + A2/A3 ? A4
? A5/A6 ? B7 ? invite.

78

79 **## 4. BETA ? polished (goal: paying pilot academies)**

80

81 Maps deliberately onto PLATFORM_ARCHITECTURE P1?P3 ? the UI roadmap and the platform
roadmap are

82 the same roadmap.

83

84 1. ****Drop 1 ? Accounts & academies (P1):**** real auth (Firebase Auth ? GCP-native, free
tier ? or

85 Clerk), academy workspace, coach invites athletes (****18+ only****, per guardrail),
roles

86 (owner/coach/athlete). Coach?athlete stays an ****optional overlay**** on the generic
schema.

87 2. ****Drop 2 ? Storage & worker (P2/P3):**** direct-to-cloud resumable uploads (signed URLs
? bypasses

88 proxy body limits entirely), reels served from object storage; control plane moves to
Cloud Run;

89 ****GPU box becomes the first pull-based `byo\_worker`**** (outbound-only, exactly P3's
security

90 model). Email notification when a reel is ready.

91 3. ****Drop 3 ? Review v2:**** keyboard shortcuts (J/K/L, [/] nudge), frame-step, timeline
minimap with

92 rally bars, batch accept, rally tags, athlete-facing ****expiring share links****, per-
player

93 aggregates and match-over-match trends (v1 stats only ? ****no gait/biometrics****, GDPR
Art. 9).

94 4. ****Drop 4 ? Monetization & consent:**** Razorpay per-match credits (INR; PMF: B2C
\$60?180/yr,

95 academy \$300?1,500/yr equivalents, tuned per-match for India), free-tier match
allowance;

96 ****training-data opt-in****: per-user toggle default OFF + per-match consent record +
adult-only

97 attestation at upload (guardrails: minors excluded from training pool; separate
explicit opt-in).

98

99 Beta effort: ~4?6 weeks part-time across the four drops; each drop independently
shippable.

```

100
101     ## 5. Decisions needed from owner
102
103     1. **Greenlight** alpha scope above (= greenlights DEFERRED_HARDENING_PLAN #16)?
104     2. **RESOLVED 2026-06-10:** product name = **KheISutra Guru**; domain =
**kheISutra.guru** (GoDaddy receipt, 3-yr registration + domain protection, renews 2029-06).
105     3. Per-user **training opt-in default OFF** ? confirm (recommended; conservative per
guardrails).
106     4. Alpha invite list: the C13 interview academies + camera-pilot contacts?
107     5. SPA stack confirm: React + Vite + Tailwind (proposed) vs anything you prefer.
108     6. **RESOLVED 2026-06-10:** separate **private repo `kheISutra`** (owner-ratified ? the
product's
109         "heart and brain"): SPA in `web/`, control plane joins at beta; backend PRs B1?B7
stay in the
110         indexer repo. Keeps the indexer's licensing/test discipline untouched.
111
112     *Once ratified, decisions 1?6 get recorded in `docs/archives/decisions/DECISIONS.md` per
convention.*
113

```

9. assistant (2026-06-21T13:56:29.844Z)

I have all four docs read. Let me verify a couple of key claims against the actual kheISutra working tree ? specifically the SPA i18n state and the site/ surface state ? to confirm what's actually shipped vs. what remains.

10. user (2026-06-21T13:56:33.504Z)

```

=== locales dir ===
ls: cannot access 'locales/': No such file or directory
=== web i18n deps ===
"@fontsource-variable/ noto-sans-devanagari": "^5.2.8",
"@fontsource-variable/ noto-sans-kannada": "^5.2.9",
"i18next": "^26.3.1",
"i18next-browser-languagedetector": "^8.2.1",
"i18next-icu": "^2.4.3",
"react-i18next": "^17.0.8"
=== site i18n markers ===
site/public/flyer-hi.html
site/public/flyer-kn.html
site/public/flyer.html
site/public/index.html
=== git log recent (kheISutra) ===
7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD $6 box 3
?) (#57)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280)
(#55)
2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it
surfaced (#54)
113587f docs: set the dry-run CUJ as a tracked goal (UI drives a bucket video -> reel download,
multilingual) (#53)
1b69e1a feat(site): show a localized UI screenshot on each academy flyer (#52)
3a42847 feat(web): localized sample-query prompt (en/kn/hi) + i18n-test rule for query parsing
(#51)
0021c4f test(web): guard the SPA stays self-contained / offline-renderable (#50)
631c3aa docs(friendly-multilingual): SPA fully trilingual ? mark B1/B2/B4/B5/B6 done +
screencast milestone (#49)
ad0d7ed feat(web): ship the Hindi (hi) catalog ? tracker B6 (the extensibility proof) (#48)
4449981 feat(web): ship the Kannada (kn) catalog ? tracker B5 (#47)
a423b82 feat(web): self-host OFL Noto Kannada + Devanagari fonts ? tracker B4 (#46)
289324e feat(web): extract all SPA strings to the en catalog + i18n parity CI ? tracker B2 (#45)
5b95b0d feat(web): i18n scaffold + language switcher (en-only) ? tracker B1 (#42)
26d636d feat(site): link the academy flyer from the top nav + own-model page (#44)

```


11. assistant (2026-06-21T13:56:38.186Z)

The git log shows work has progressed beyond what the doc's \$7 tracker records. There are PRs #49-#57 that aren't reflected in the tracker rows I read. Notably: `locales/` is NOT at repo root (the doc's design assumed it there) ? it must live under `web/`. And there are new flyer-kn/flyer-hi pages, plus a sample-query prompt (#51, which is tracker A9-related). Let me check where locales actually live and the state of the site/ surface and the newer work.

12. user (2026-06-21T13:56:44.125Z)

```
=== find locales ===
./web/src/i18n/locales
=== web/src/locales ===
web/src/i18n/locales/en/common.json
web/src/i18n/locales/hi/common.json
web/src/i18n/locales/kn/common.json
=== site/ i18n in server.js or shim ===
=== whatsapp share present? ===
site/public/flyer-hi.html
site/public/flyer.html
=== A-items: error mapping / touch targets markers ===
```

13. assistant (2026-06-21T13:56:48.253Z)

Confirmed the key facts. Let me verify the doc's \$7 tracker is stale relative to git (PRs #49-#57 not reflected) and check what the newer PRs actually did regarding A-items, since the tracker only goes up to #48-ish in the rows but git shows more. Let me check the flyer localization and the sample-query work, plus confirm site/ still has no shim.

14. user (2026-06-21T13:56:53.345Z)

```
=== site/public html files ===
site/public/capture.html
site/public/flyer-hi.html
site/public/flyer-kn.html
site/public/flyer.html
site/public/index.html
site/public/own-model.html
=== flyer-kn/hi are full localized pages? (lang attr) ===
<html lang="hi"
<html lang="kn"
<html lang="en"
=== index.html still en-only (no kn/hi switch)? ===
1
=== capture.html localized? ===
0
=== A9 sample-query in web ===
=== does tracker doc mention #49-#57? ===
0
```

15. assistant (2026-06-21T13:57:01.098Z)

The picture is clear and well-verified. The `index.html` has 1 match (let me confirm it's not a real switcher ? likely a stray), and capture.html is fully English. The site/ marketing surface (B3) still has no `t()` shim or `data-i18n` ? the only localized site pages are the standalone hand-built flyer-kn/flyer-hi.html (academy flyers from #44/#52, not the main funnel pages). The \$7 tracker doc doesn't reference PRs #49-#57, so it's mildly stale (newer work like the localized sample-query #51 partially advances A9; flyer localization #52 is adjacent but not a tracker row).

Let me do one final check on that single index.html match and confirm the engine P2 / B7 status.

16. user (2026-06-21T13:57:05.143Z)

```

=== the 1 index.html match ===
589:      <div style="margin-top:.4rem">Academy flyer (print): <a href="/flyer"
hreflang="en">English</a> . <a href="/flyer-kn" hreflang="kn" lang="kn">?????</a> . <a
href="/flyer-hi" hreflang="hi" lang="hi">?????</a></div>
=== flyer link from index nav (the localized funnel entry) ===
=== ENGINE P2/backend i18n present? ===
find: 'backend': No such file or directory
ls: cannot access 'backend/i18n': No such file or directory
grep: backend/main.py: No such file or directory
grep: backend/reporting/spec.yaml: No such file or directory

```

17. assistant (2026-06-21T13:57:09.012Z)

Confirmed. The single index.html "match" is just the flyer download links (English/Kannada/Hindi print flyers) ? not a site-content language switcher. The site's main funnel pages (index.html, capture.html, own-model.html) remain English-only with no `t()` shim/data-i18n (B3 not started). Engine `backend/` isn't in khelsutra (it's the indexer repo) ? B7 is correctly cross-repo and unstarted there.

I have everything needed. Let me compile the structured findings.

18. user (2026-06-21T13:57:56.144Z)

Structured output provided successfully

19. assistant (2026-06-21T13:58:08.105Z)

I've completed the localization + UX track review. Structured findings returned via StructuredOutput. Summary of what I verified against the actual khelsutra working tree (not just the doc's claims):

****Track home:**** `C:/Users/avidu/Projects/khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md` (\$7 = source of truth).

****DONE (?)**** P0, B1, B2, B4 (OFL Noto fonts), B5 (kn), B6 (hi) ? the `web/` SPA is fully trilingual en+kn+hi. Confirmed: `web/package.json` has i18next/react-i18next/i18next-icu/langdetector + @fontsource Noto Kannada/Devanagari; catalogs at `web/src/i18n/locales/{en,kn,hi}/common.json`. 4 of 5 owner gates locked (D6?D9).

****What REMAINS:****

- ****B3**** (site/ `t()` shim + data-i18n) ? biggest remaining row; verified the site funnel pages (index/capture/own-model.html) are still English-only. The only localized site assets are standalone print flyers (flyer-kn/hi.html), not the funnel.
- ****A-items**** A1 (de-jargon copy), A2 (WhatsApp share ? none wired today), A4 (mobile touch targets), A5 (friendly error mapping), A6, A7, A8, A9 (tap-only filters ? partially advanced by #51's localized sample-query prompt). All ungated except via deps.
- ****A3**** (recording-education kn/hi) ? ? blocked on B3.
- ****Native-speaker review**** of AI-drafted kn/hi ? owner-gated.
- ****Engine P2 / B7**** (backend localization) ? ? owner-gated, cross-repo; confirmed no `backend/i18n` in the engine repo.

****Flagged staleness (did not fix):****

1. \$7 tracker stops at ~#48-49 but git shows #51 (advances A9) and #52/#44 (flyer localization) unrecorded.
2. Catalogs live under `web/src/i18n/locales/`, not the root-level shared `locales/` the design (D-CATALOG-HOME) diagrams ? so B3's "reuse the SAME catalogs across surfaces" premise isn't yet realized.
3. I18N_PLAN.md \$3/\$5 still name the retired `backend/api/static` as P1's target (reconciled only via the top banner).